

Wrist Stability Guard



Prototype by: Torian M. Stiles

Prototype Overview

What it Does

This prototype is a speculative creation of a specialized hand guard that stabilizes the wrist. Since the 1st iteration of the prototype, it used padding to serve as a hand rest that makes it easier for people to use keyboards and access the internet. The prototype features a wearable wrist guard and glove that was designed with the intention of aiding stability, with comfort and accessibility being the biggest priority throughout this process. It acts as both a glove or fingerless glove depending on the model while supporting its primary usage cases.

Target Audience

This project is dedicated to supporting those with arthritis for the goal of interacting with computer keyboards, and protecting their body by minimizing pain in the wrist area. Any person is allowed to experiment with the two models, but the augmentation is intent on supporting those with arthritis and general wrist pain. For people with other hand-related conditions, it is best to avoid this prototype's concept unless it directly enhances or supports your hand without repercussions.

What it Augments

The prototype augments the ability to interact with technology (specifically keyboards) to access the internet. Those who feel hand or wrist pain benefit from the ability to rest them on a surface and use a computer without much issue. The padding inside of the glove, and stabilization offered by the wrap, both improve the user's experience with technological accessibility. It acts as a supportive brace similar to those worn at a hospital but with the freedom of using your fingers for keyboard typing or moving a mouse.

The Iterative Process

This prototype took many forms, one of which being a speculative drawing that captures the look of a fingerless glove that had loop rings that attach to the top of the wrist guard. This was a design choice that did not make it to the final cut for potential risks, however the general protective brace remained a constant variable. It started as a drawing, then became visualized using a powerful 3D modeler AI called printpal to vibe-code this idea into an interactable, digital design. The choice to move to a more effective design than a cool one was a choice made with the user's benefit in mind. A lot of variants were produced that did not capture the look or feel, that had changed after revising the main prompt to get to a positive outcome.

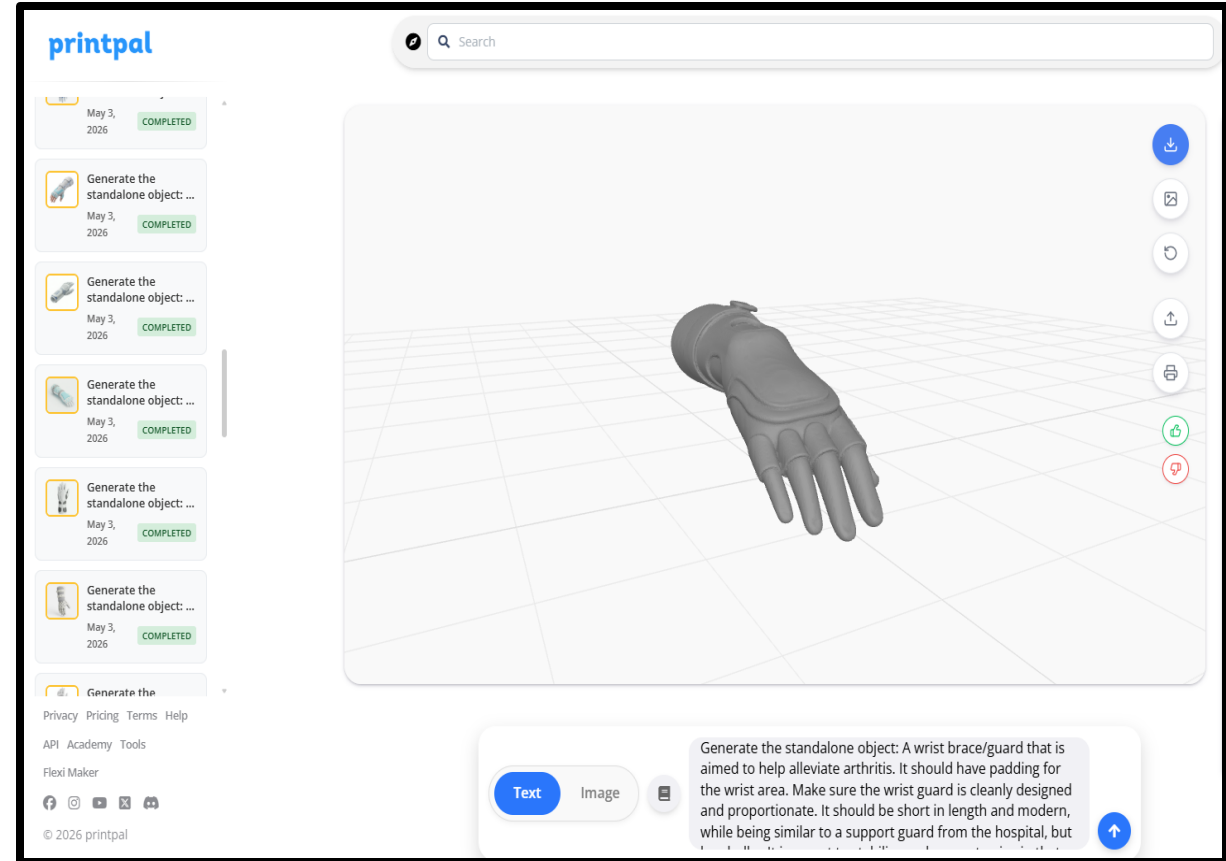


Model Creation

Prompt:

“Generate the standalone object: **A wrist brace/guard** that is aimed **to help alleviate arthritis**. It should have padding for the wrist area. Make sure the wrist guard is **cleanly designed and proportionate**. It should be **short in length** and modern, while being similar to a support guard from the hospital, but **less bulky**. It is meant to stabilize and prevent pains in that area. Go for a **simple look** if possible. Ensure that it accounts for **padding and stabilization**. Make it more like a glove as well as a protective guard. Take every detail into account. Keep the design clean and smooth, as it is an augmentation asset. Since it is not a long object, ensure after it cuts off, the prototype is just the **combination of a wrist guard and glove**. Stylize the 3D mesh to make it more effective and while keeping this new design very consistent, proportionally accurate, and **parallel in its features**. It should represent a model of a fingerless glove, hollow shell, open fingertips, no finger caps, open hollow inside, high-fidelity texture and optimized.”

Note: The software produced two variants out of 36 that serve the intended purpose.



Interact with Model 1!



Link: [Prototype Model 1](#)

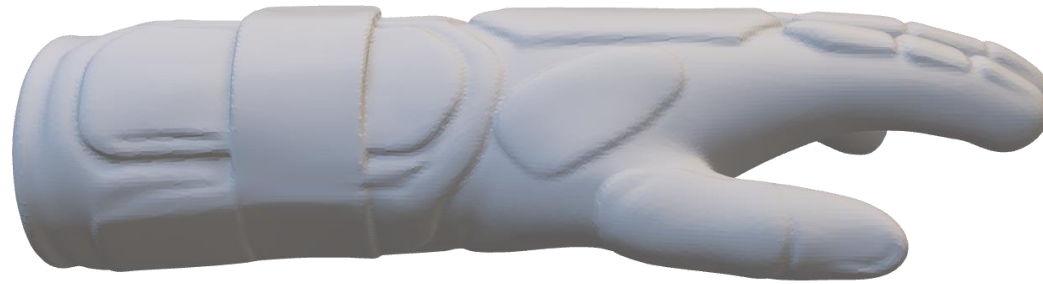
Constraints

One of the biggest issues with making the prototype was figuring out how to best represent it. Starting as a drawing, the design of this prototype was dictated by software capabilities and my own imagination. The drawing moved to an idea of being a model in TinkerCad, but producing a such a design with limited experience and time available was what led to where it stands today. I knew I made the right choice with leveraging the modern abilities of printpal's artificial intelligence, and paired it with innovative prompting. One other constraint however, involved the creation of two models but that serves as a positive for being more versatile and possibly, one design may work better than the other while accomplishing the same goal. A final constraint I faced when it came to creating the designs were the AI's generative versions which failed a lot of times and required more prompt refinement and luck to get a effective style and appearance. More often than not, it came down to patching errors made by the program itself and tinkering with words to get a good result for the way it worked and would look. I did have to make due with the current look of these models, but found their features to be a great choice in the end.

Main Issues

- Time
- Complexity
- AI

Interact with Model 2!



Link: [Prototype Model 2](#)

Competencies

The design process taught me a lot of skills I did not know I had, while improving upon others. I learned to craft this design with a user-first approach, focusing on the overall efficiency and user experience that the prototype offers to the target group of this augmentation. The ability to think creatively is a skill that gave me the competency of creative thinking, while also demonstrating the competency of iterative design. AI and vibe-coding became an efficient way to bring my design to life, proving that my capabilities were fundamental skills in the process. The goal of helping those with arthritis and similar hand/wrist issues to access the internet was informed by these competencies and the willingness to inspire new ideas. My course in TCID (Tech Writing and Design) 4060 course helped me develop these skills as I started from the initial phase of the prototype to the end product of it being digitally molded and interactive. The main competencies derived from concepts of accessibility, AI collaboration, and technological augmentation that introduced my brain to bringing an idea to fruition in a creative way. It was a team effort between all three of these main competencies that formed the two versions of the wrist guard, with the goal being to use my design as theory and application for future accessibility practices.

Main Competencies

- Creative Thinking
- Artificial Intelligence (AI) Vibe Coding
- Iterative Design

Conclusion

Making this augmentation became a valuable lesson for understanding accessible design and the user experience. By crafting a prototype of a wrist guard that protects and stabilizes the hand for using keyboards, I learned that the prototype's objective was primarily user focused and improved their accessibility with using technology. The design starting as an idea, becoming a paper prototype, then becoming high-fidelity models gave me insight to the user experience as I considered new possibilities and arguments for their prompt with printpal. I had to really consider the efficiency and how a person would safely wear these augmentations without pains or inhibition of finger movement. I gained a new skill in design accessibility; being able to think for the betterment of others at the cost of something else. The cost of this however, was the ability to use finger rings that tied to the top for stability of the fingers, which became good to throw out to avoid potential danger and improve overall stabilization and hand typing. The next phase of this augmentation is to make it real, and possibly better the lives of many of us who are aging or suffering from related conditions.

This prototype should serve as a reminder that we can improve not only the user experience, but the accessibility of our lives from just a thought.

Questions and Feedback

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I respond in under 24 hours!

Feel free to discuss, ask questions, or give input about this design.